# Dohyun Lee

Ph.D. Student in Artificial Intelligence

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## About

My current research interest lies in how robots can build and use an understanding of the 3D world to manipulate objects and act in real environments. My research centers on two directions. First, while building PUBG Ally at KRAFTON, I saw first-hand that vision-language models understand complex 3D scenes remarkably well, and I now aim to bring this capability to robot manipulation, together with tools such as SAM 3. Second, video is another rich source of 3D world understanding. I believe that video generation is essentially representation learning for the 3D world and its dynamics, and I am currently studying what learning objectives yield good representations for world action models.

I am currently a Ph.D. student at the DAVIAN Lab, KAIST AI, advised by Prof. Jaegul Choo. Previously, I was a research intern at KRAFTON.

## Education

**Korea Advanced Institute of Science and Technology** — *Seongnam, S.Korea*

Integrated M.S./Ph.D. in Artificial Intelligence (Advisor: Prof. Jaegul Choo, DAVIAN Lab) — *Sep. 2023 - Feb. 2029 (Expected)*

**Kyunghee University** — *Yongin, S.Korea*

B.S. in Mechanical Engineering and Software Engineering — *Mar. 2017 - Aug. 2023*

- GPA: 4.27/4.5 (Major: 4.29/4.5)
- 4 Academic Excellence Scholarship, 2 Academic Excellence Award

## Research Experience

**KRAFTON** — *Seoul, S.Korea*

Research Intern, AI Companion Team — *Nov. 2024 - Feb. 2026*

- Researched and developed PUBG Ally, showcased at CES 2025 in collaboration with NVIDIA (LINK).
- PUBG Ally was beta-launched in the live game PUBG: BATTLEGROUNDS (LINK).

**Hyundai Motor Company** — *Seoul, S.Korea*

Undergraduate NLP Researcher — *Jul. 2022 - Aug. 2022*

## Publications

**Soft-Value Flow Matching for Offline Reinforcement Learning**

**D. Lee**[*], J. Hong[*], T. Choi, K. Kim, J. Choo, K. Lee, Y. Oh — *2026*

Preprint, 2026

**PointmapVLA: Scaling Robot Learning with Robot-Centric Pointmaps**

D. Hwang[*], D. Kim[*], B. Lee[*], K. Lee, **D. Lee**, H. Lee, M. Park, J. Choo — *2026*

Preprint, 2026

**An Agentic Framework for Proactive Robots with Procedural Assistance Memory**

**D. Lee**[*], M. Park[*], J. Park, D. Hwang, J. Hyung, S. Hwang, B. Lee, J. Choo — *2026*

Conference on Robot Learning (CoRL), 2026

**PersonaTrail: Benchmarking Personalized Web Agents through Browsing Trails**

S. Yang, C. Ki, **D. Lee**, J. Choo, C. Park — *2026*

arXiv preprint, 2026

**Delving into Large Language Models for Effective Time-Series Anomaly Detection**

J. Park, K. Jung, **D. Lee**, H. Lee, D. Gwak, C. Park, J. Choo, J. Cho — *2025*

Advances in Neural Information Processing Systems (NeurIPS), 2025

**Opt-Out: Investigating Entity-Level Unlearning for Large Language Models via Optimal Transport**

M. Choi, D. Rim, **D. Lee**, J. Choo — *2025*

Annual Meeting of the Association for Computational Linguistics (ACL), 2025

**Revisiting LLMs as Zero-Shot Time Series Forecasters: Small Noise Can Break Large Models**

J. Park, H. Lee, **D. Lee**, D. Gwak, J. Choo *2025*

Annual Meeting of the Association for Computational Linguistics (ACL), 2025

**Exploring In-context Example Generation for Machine Translation**

**D. Lee**, S. Lee, C. Yang, Y. Baek, J. Choo *2025*

Findings of the Association for Computational Linguistics (ACL), 2025

**Breaking Chains: Unraveling the Links in Multi-Hop Knowledge Unlearning**

M. Choi, C. Park, **D. Lee**, J. Choo *2024*

arXiv preprint, 2024

**PairEval: Open-domain Dialogue Evaluation with Pairwise Comparison**

C. Park, M. Choi, **D. Lee**, J. Choo *2024*

Conference on Language Modeling (COLM), 2024

**Protecting Privacy Through Approximating Optimal Parameters for Sequence Unlearning in Language Models**

**D. Lee**[*], D. Rim[*], M. Choi, J. Choo *2024*

Findings of the Association for Computational Linguistics (ACL), 2024